

Hyunjin Choi (최현진)

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EDUCATION

Sogang University Ph.D. Department of Mechanical Engineering Dissertation Title: Assistance and Rehabilitation of People with Paretic Gait Using a Torque-controlled Wearable Robot Advisor: Kyoungchul Kong	2019
Sogang University B.S. Department of Physics (Magna Cum Laude) B.E. Department of Economics	2012

PROFESSIONAL EXPERIENCE

Sangmyung University Assistant Professor Department of Human Intelligence Robot Engineering	2020 - present
Angel Robotics Co., Ltd Founding Member / Lead Engineer	2017 - 2020
University Paris-Est Créteil (UPEC) Visiting Researcher Laboratoire Images, Signaux et Systèmes Intelligents (LiSSi)	Summer 2013, 2014, 2015
Sogang University (Industry-University Cooperation Foundation) Student Researcher EXO Lab (formerly RSC Lab)	2012 - 2016

ACADEMIC SERVICE

IFAC Technical Committee on Robotics (TC 4.3) Vice Chair	2024 - Present
ICROS (Institute of Control, Robotics, and Systems) Organizing Committee Member	2024 - 2026
IFAC Symposium on Robotics Proceedings Editor	2025
IFAC World Congress Associate Editor	2026

FELLOWSHIPS AND AWARDS

Gold and Bronze Medal in Powered Exoskeleton Race Cybathlon Team Angel Robotics	2020
Young Researcher Award (우수신진연구자상) Institute of Control, Robotics and Systems (제어로봇시스템학회)	2019
Top 10 Student Project Worldwide National Instruments	2017
Top 10 Finalist UAE AI & Robotics for Good Project: Angelegs	2016
Global Ph.D. Fellowship National Research Foundation (NRF), Korea Project: Study on a mechatronic rehabilitation system incorporating functional electrical stimulation and a robotic exoskeleton	2012 - 2016

† indicates corresponding author(s)

Journal Papers

1. H. Kim, K. Shi, **H. Choi**, D. Rha, K. Kong[†], “Real-time Spatiotemporal Gait Analysis Using a Hip Assistive Wearable Robot,” *IEEE Access*, vol. 14, pp. 36025–36035, 2026.
2. D.-E. Lim, S. Koo, **H. Choi**[†], “Hybrid Physics–LSTM Model for Vision-Based Ball Landing Point Prediction,” *Journal of the Korea Institute of Information and Communication Engineering (JKIECE)*, vol. 30, no. 2, pp. 217–223, 2026.
3. S. Koo, J.-H. Jo, **H. Choi**[†], “Initial-contact-timing Estimation Based on Supervised Learning in a Hip-assistive Wearable Robot,” *Journal of Institute of Control, Robotics and Systems*, vol. 32, no. 2, pp. 158–163, 2026.
4. J.-H. Jo, S. Koo, K.-H. Min, **H. Choi**[†], “Development of an FSR-IMU Integrated Insole System for Gait Analysis,” *The Journal of The Korea Institute of Electronic Communication Sciences (KIECS)*, vol. 21, no. 1, pp. 391–397, 2026.
5. S. Koo, J.-H. Jo, K.-H. Min, **H. Choi**[†], “Research on a Wearable Hand Gesture Recognition System Based on Pressure Sensors,” *The Journal of The Korea Institute of Electronic Communication Sciences (KIECS)*, vol. 20, no. 6, pp. 1367–1373, 2025.
6. H.-c. Kim, **H. Choi**[†], “Design of a Shoe-Mounted Ground Inclination Measurement System Using Time of Flight Sensors,” *The Journal of The Korea Institute of Electronic Communication Sciences (KIECS)*, vol. 19, no. 5, pp. 1005–1012, 2024.
7. H.-c. Kim, **H. Choi**[†], “Heel Trajectory Analysis Method of Walking using a Wearable Sensor,” *The Journal of The Korea Institute of Electronic Communication Sciences (KIECS)*, vol. 18, no. 4, pp. 731–736, 2023.
8. S. An, J. Feng, E. Song, K. Kong, J. Kim, **H. Choi**[†], “High-Accuracy Hand Gesture Recognition on the Wrist Tendon Group Using Pneumatic Mechanomyography (pMMG),” *IEEE Transactions on Industrial Informatics (TII)*, vol. 20, no. 2, pp. 1550–1561, 2023.
9. J.-h. Seong, **H. Choi**[†], “A Wearable Glove System for Rehabilitation of Finger Injured Patients,” *The Journal of The Korea Institute of Electronic Communication Sciences (KIECS)*, vol. 18, no. 2, pp. 379–385, 2023.
10. J. Choi, *et al.*, **H. Choi**, K. Kong[†], “The History and Future of the WalkON Suit: A Powered Exoskeleton for People With Disabilities,” *IEEE Industrial Electronics Magazine*, 2022.
11. H. Choi, J. Yeo, J.-h. Seong, **H. Choi**[†], “Development of Low-cost 3D Printing Bi-axial Pressure Sensor,” *Journal of Korea Robotics Society*, vol. 17, no. 2, pp. 152–158, 2022.
12. **H. Choi**, “Assistance of a Person with Muscular Weakness Using a Joint-Torque-Assisting Exoskeletal Robot,” *MDPI Applied Sciences*, vol. 11, no. 7, p. 3114, 2021.
13. **H. Choi**, K. Kong[†], “A Soft Three-Axis Force Sensor Based on Radially Symmetric Pneumatic Chambers,” *IEEE Sensors Journal*, vol. 19, no. 13, p. 5229, 2019.
14. **H. Choi**, K. Kong[†], “A User Interface System with See-Through Display for WalkON Suit: A Powered Exoskeleton for Complete Paraplegics,” *MDPI Applied Sciences*, vol. 2, p. 2287, 2018.
15. **H. Choi**, S. Oh, K. Kong[†], “Control of a Robotic Manipulator in the Polar Coordinate System using a Biarticular Actuation Mechanism,” *International Journal of Control, Automation, and Systems (IJCAS)*, vol. 14, no. 4, pp. 1095–1105, 2016.
16. B. Na, **H. Choi**, K. Kong[†], “Design of a Direct-Driven Linear Actuator for a High-Speed Quadruped Robot, Cheetaroid-I,” *IEEE/ASME Transactions on Mechatronics*, vol. 20, no. 2, pp. 924–933, 2014.

Refereed Conference Papers

1. 장우혁, 정민규, 최승민, 강범수, **최현진**[†], “Development of a Real-time Exercise Posture Estimation System using a Wearable Multi-modal Sensor Network,” *ICROS 2026*.
2. S. Koo, W. Jang, M.-g. Jung, **H. Choi**[†], “Gait-Period-Informed Bi-LSTM Estimation of Knee Kinematic Peaks from Hip Joint Kinematics,” *ICROS 2026*.
3. 임재승, 이상현, 김민서, **최현진**[†], “Development of a Vision-based Elevator Door Control System for Boarding Safety of Mobility-Vulnerable Users,” *ICROS 2026*.
4. 홍성은, 박병선, 전서진, **최현진**[†], “Design of a Wrist Posture-Based Carpal Tunnel Syndrome Prevention System,” *ICROS 2026*.
5. 나재웅, 김민섭, 김도균, **최현진**[†], “Development of a Wearable Bidirectional Communication Assistance System for the Hearing Impaired Using Vosk and MediaPipe,” *ICROS 2026*.
6. 임승현, 김현민, 고동진, **최현진**[†], “An adaptive robotic gripper system based on real-time object recognition,” *ICROS 2026*.

7. 노재윤, 임태운, 이나현, **최현진**[†], “Cognitive-Based Adaptive Camouflage robot Platform for Prior Reconnaissance in Dangerous Areas,” *ICROS 2026*.
8. S. Koo, J.-H. Jo, **H. Choi**[†], “Abnormal Gait Detection in Parkinson’s Disease Using a Waist-Worn Robot,” *KRoC 2026*.
9. C. Ko, K. Kong, **H. Choi**[†], “Real-time Gait Recognition of Multiple Stair-climbing Modes for Hip-assisting Wearable Robots,” *IFAC Symposium on Robotics 2025*.
10. S. Koo, K.-H. Min, J.-H. Jo, **H. Choi**[†], “A Study on Hand Gesture Recognition System Using a Wearable Pressure Sensor Band,” *ICROS 2025*. (학부생논문상)
11. 하성민, 김선우, 정세림, **최현진**[†], “Door Element Detection and Distance Measurement System for the Visually Impaired Using Stereo Vision and YOLOv8 Early Fusion,” *ICROS 2025*.
12. 문태식, 이수민, 임동혁, **최현진**[†], “A MediaPipe-Based Sign Language Translation System for Improving Hospital Medical Services,” *ICROS 2025*.
13. J.-H. Jo, K.-H. Min, S. Koo, **H. Choi**[†], “Gait Analysis Insole Research Based on Pressure Sensing Sensor and Inertial Measurement Technology,” *ICROS 2025*.
14. 이준우, 조훈희, 이예진, **최현진**[†], “A Posture-Adaptive Shield Control Robot System Based on 3D LiDAR and Infrared Keypoint Detection,” *ICROS 2025*. (학부생논문상)
15. 임다은, 강현민, 권수빈, **최현진**[†], “Landing Point Prediction of a Freely Moving Ball Using a Vision-Based Hybrid Model,” *ICROS 2025*.
16. H.-c. Kim, J. Kim, E. Song, **H. Choi**[†], K. Kong, “Feasibility of pMMG Sensors for Muscle Activity Assessment in Squat Exercises,” *ICCAS 2024*.
17. 최정빈, 신승연, 이성현, 최상균, **최현진**[†], “Quantity-Adjustable Meal Assistance Robot System,” *ICROS 2024*. (학부생논문상)
18. 이종혁, 이유빈, 김유중, **최현진**[†], “Development of Remote Indoor Security Platform Based on Visual-Audio Information,” *ICROS 2024*.
19. 노승서, 김영준, 윤정빈, **최현진**[†], “Deep Learning Based Development of a Web Service for identifying Drug-Food Interactions,” *ICROS 2024*. (학부생논문상)
20. 남택권, 자호라 바툴 제프리, 지태규, 송지호, **최현진**[†], “Generative AI-Based Intelligent Vehicle Proximity Space Perception System,” *ICROS 2024*.
21. 김준수, 조영빈, **최현진**[†], “Development of a Wearable Fire Alarm System for the Hearing Impaired,” *ICROS 2024*.
22. T. Kim, E. Song, S. An, **H. Choi**[†], K. Kong, “Leaf-Inspired FSR Array and Insole-Type Sensor Module for Mobile Three-Dimensional Ground Reaction Force Estimation,” *ICRA 2024*.
23. E. Song, S. An, M. Kang, **H. Choi**[†], K. Kong, “Wearable Robot Embedded Human Muscle Activity Monitoring System and Human-Oriented Joint Torque Calculation,” *IFAC World Congress 2023*.
24. 황윤미, 김지민, 조경근, **최현진**[†], “Device for Fall Detection System for Elderly Living Alone,” *ICROS 2023*. (학부생논문상)
25. 박주형, 김영운, 황유성, 김동준, **최현진**[†], “Automatic Trash Sorting System Using Deep Learning,” *ICROS 2023*.
26. 안성우, 박세민, 김재현, 김동규, **최현진**[†], “Ceiling Type Automatic Serving Robot Using Object Recognition and Image Processing,” *ICROS 2023*.
27. 최근영, 조현근, 하태형, 장호렬, **최현진**[†], “A Finger Wearable Device that Reads Braille for The Visually Impaired,” *ICROS 2023*. (학부생논문상)
28. 조찬우, 문정원, 이지훈, 김은찬, **최현진**[†], “Design and Construction of a Low-Cost Home Smart Security System Capable of Real-Time Monitoring,” *ICROS 2023*.
29. 김효원, 구혜진, 김나빈, **최현진**[†], “Customized Information Service Using Face Recognition,” *ICROS 2023*.
30. 이소화, 석혜원, 김예지, 신유진, **최현진**[†], “Ankle Rehabilitation Assistive Robot for Patients with Lateral Collateral Ligament Injuries,” *ICROS 2023*. (학부생논문상)
31. 구유하, 김희찬, 문건수, **최현진**[†], “A Gait Phase Prediction System using Wearable Sensors,” *ICROS 2023*.
32. 박지훈, 이우진, 이찬진, **최현진**[†], “Development of Pothole Detection Algorithm using 3D point Cloud,” *ICROS 2023*.
33. 이성민, 권시은, 박은서, 전동렬, **최현진**[†], “Posture Correction System Using Artificial Intelligence,” *ICROS 2023*.
34. S. An, E. Song, **H. Choi**[†], K. Kong, “Measurement of Force Myography in Level Gait and Stair Gait,” *IFAC Symposium on Mechatronics 2022*.
35. E. Song, S. An, **H. Choi**[†], K. Kong, “Soft Sensor Unit for Measurement of Large Scale Multi-axis Force,” *Asian Control Conference (ASCC 2022)*.
36. 김민경, **최현진**[†], “Posture Analysis and Classification System Using Wearable Sensors and k-NN,” *ICROS*

2022.

37. 유진영, 김정현, 김현준, 방유석, **최현진[†]**, “A Study on the Height Adjustable and Table Setting Serving Robot,” *ICROS 2022*.
38. 여준성, 최현수, 성지훈, **최현진[†]**, “Design of Soft Force Sensor with Gallium-Filled Microchannels,” *ICROS 2022*. (학부생논문상)
39. 김세호, 모시원, 김경수, 김영재, **최현진[†]**, “Customized Haptic Gloves for Better User Experience,” *ICROS 2022*. (학부생논문상)
40. 안수빈, 한상현, 정재혁, 방시온, 김지현, **최현진[†]**, “HandShape Recognition Display for Infant Small Muscle Development,” *ICROS 2022*.
41. 정성호, 김재욱, 김희찬, 최현수, **최현진[†]**, “Mitigating side effects of crutches and walking assistance smart crutches device,” *ICROS 2022*.
42. 여준성, 김범석, 허광성, 현무림, **최현진[†]**, “Industrial Disaster Prevention Smart Safety Equipment Using ZigBee and central safety management system,” *ICROS 2022*.
43. 남재욱, 김민경, 김성연, 이상우, **최현진[†]**, “Ceiling Electric Vehicle Charge System Using Magnetic Connector,” *ICROS 2022*.
44. 유동현, 김동규, 서동민, 전우혁, **최현진[†]**, “Development of Text to Braille Translate Device Using OCR,” *ICROS 2022*.
45. 최선일, 김규린, 정태혁, 신희준, 황병욱, **최현진[†]**, “Modular Autonomous Parking System for Shared Scooters,” *ICROS 2022*.
46. S. An, **H. Choi**, K. Kong[†], “Development of Wireless Pneumatic Myography Sensor for Real-time Muscle Contraction Measurement,” *ICCAS 2021*.
47. 전대현, 김시원, 성지훈, 조민지, **최현진[†]**, “A Research on the Performance of Force Feedback Using V/AR,” *ICROS 2021*.
48. 김보경, 김지현, 최혜정, **최현진[†]**, “Foot Pressure and Movement Measurement Analysis For Standing State Detection,” *ICROS 2021*.

COURSES TAUGHT AT SANGMYUNG UNIVERSITY

Signals and Systems	Fall 2020, 2021, 2022, 2023
Robot Operating System (ROS)	Fall 2020, 2021, 2022, 2023, 2024
Human-Robot Interface Engineering	Fall 2020, 2021, 2022, 2023
Control Engineering and Design	Spring 2021, 2022, 2023
Capstone Design I	Spring 2021, 2022, 2023, 2024
Probability and Statistics	Spring 2021, 2022
Healthcare and Rehabilitation Robotics	Spring 2023, 2024

RESEARCH GRANTS

Current

Development of a wearable robot platform for assisting elderly individuals and paralysis patients using ultra-lightweight, high-power hybrid actuator based on shape memory alloy-coil fiber artificial muscles 2025 – 2029

Co-PI · National Research Foundation of Korea

(형상기억합금 코일 섬유형 인공근육기반 초경량·고출력 하이브리드 구동기를 적용한 고령자 및 하지마비 장애인 보조용 웨어러블 로봇 플랫폼 개발)

Past

A Study on the Built-in Motion Analysis System of Wearable Gait Rehabilitation Robot 2021 – 2024

PI · National Research Foundation of Korea

(착용형 보행재활로봇의 내장형 동작분석 시스템 연구)

PATENTS

Motion Intention Measurement Sensor and Motion Intention Measurement System	Korean Patent No. 10-2841080 (Jul. 2025)
Overshoe-type device for measuring ground reaction force	Korean Patent No. 10-2219037 (Feb. 2021)
Driving unit of multi-joint robot	Korean Patent No. 10-2244673 (Apr. 2021)
Controlling method of lower body assisting robot	Korean Patent No. 10-2162485 (Sep. 2020)
Ground reaction force sensor module	Korean Patent No. 10-2095439 (Mar. 2020)
Multi-axis force sensor and multi-axis force sensing system	Korean Patent No. 10-2041710 (Oct. 2019)

Crutch for wearable robot system
Method of using wearable assisting device

Korean Patent Pending No. 10-2022-0064912 (2022)
Korean Patent Pending No. 10-2022-0014421 (2022)